

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ac1dec1de83693ad8.jsonl`  
- SHA-256: `2d49083fd38783a9fdfe896a55fe46a7da38bd8bc06f12a29c6732dcbb2b3ac8`  
- Source modified: `2026-06-06T17:35:17+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:34:12.056Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"The acoustic ML classifier achieves >95% detection accuracy, but the end-to-end on-court system drops to 85% average accuracy, showing real-world deployment degrades audio-only hit detection substantially."

****Source:**** <https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/> (primary)
****Supporting quote:**** "the ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:34:16.874Z)

I'll verify this claim by examining the source and checking whether the quote supports it.

3. assistant (2026-06-06T17:34:19.776Z)

I'll fetch the primary source to verify the exact quote and context.

4. user (2026-06-06T17:34:28.039Z)

Tennis Timing Detection System - Accuracy Analysis

****Sport:**** Tennis

****What's Being Detected:**** Ball impacts on rackets and ground rebounds during tennis rallies and groundstrokes

****System Type:**** Audio-based acoustic detection using machine learning

****System Components:**** Commercial microphone (built into action camera), notebook computer, ML algorithm

Accuracy Statements

****ML Algorithm Detection:****

The document states: "The ML algorithm showed a detection accuracy higher than 95%" when classifying sounds in controlled lab settings.

****On-Court System Performance:****

The overall system accuracy decreased in real-world conditions: "the average accuracy of the whole system that was applied on-court was 85%."

Context for the Discrepancy

The higher lab accuracy (>95%) applies specifically to the machine learning classifier's ability to distinguish impact and rebound sounds from noise during isolated testing. The lower on-court accuracy (85%) reflects the complete system's performance in actual tennis environments with ambient noise, wind, and other interference. The system detected impacts at 96.25%-81.25% accuracy depending on player group, but rebound detection was lower at 86.25%-70.83%.

5. assistant (2026-06-06T17:34:31.596Z)

The source confirms the quote, but I need to dig deeper into context. The claim says "audio-only hit detection" degrades to 85%. Let me check what the "85% whole system" actually measures - it might include rebound/ground bounce detection (not just hits), and the sport is tennis, not badminton. Let me examine more carefully.

6. user (2026-06-06T17:34:40.039Z)

Tennis Timing Detection System Analysis

System Components & Goals

The system performs multiple tasks beyond ML classification:

1. **Peak detection** (audio signal processing)
2. **ML classification** (distinguishing tennis sounds from noise)
3. **Timing calculation** (measuring intervals between events)

The stated objective: measuring rally rhythm (shots per minute) and groundstroke execution timing (time between ball bounce and racket impact).

Accuracy Drop (95% ? 85%)

The abstract states: "The ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%."

This drop occurs because:

- **ML model alone** achieves 95%+ on controlled datasets
- **On-court system** integrates peak detection + ML classification + environmental noise filtering, which compounds errors across stages
- Rebound detection particularly suffers (70.83% in one group) due to low-amplitude sounds masked by ambient noise

Participant Groups

The study involved **10 total participants** across two protocols:

- **Rally assessment**: 4 male players (ranked second category by Italian Tennis Federation), ages 21.1 ± 3.7 years
- **Groundstrokes**: 6 participants (3 female, 3 male), ages 40.8 ± 13.6 years, divided into experience groups

All were amateurs or lower-ranked semi-professional players?not elite professionals.

7. assistant (2026-06-06T17:34:43.177Z)

I now have a clear picture. Let me assess the claim carefully against what the source actually says, and do a quick search for any contradicting/qualifying context on audio hit detection accuracy.

8. user (2026-06-06T17:34:50.303Z)

Web search results for query: "audio-based shot hit detection accuracy racket sports tennis badminton on-court real-world"

Links: [{"title": "Wearable Audio and IMU Based Shot Detection in Racquet Sports", "url": "https://arxiv.org/pdf/1805.05456"}, {"title": "[1805.05456] Wearable Audio and IMU Based Shot Detection in Racquet Sports", "url": "https://arxiv.org/abs/1805.05456"}, {"title": "Wearable Audio and IMU Based Shot Detection in Racquet Sports | Request PDF", "url": "https://www.researchgate.net/publication/325168095_Wearable_Audio_and_IMU_Based_Shot_Detection_in_Racquet_Sports"}, {"title": "(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis", "url": "https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"}, {"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}]

I've found several relevant research papers on audio-based shot hit detection in racket sports. Here are the key findings:

Audio-Based Detection Accuracy

The machine learning algorithm for audio-based detection showed accuracy higher than 95%, while the average accuracy of the whole system when applied on-court was 85% in table tennis studies.

Wearable Audio and IMU Fusion

A real-time system for shot detection in table tennis based on fusion of IMU and audio sensor data embedded in a wrist-worn wearable achieved 95.6% accuracy.

Multi-Sport Coverage

The research spans multiple racket sports:

- **Tennis**: An embedded device in a tennis racket handle detected tennis shots with 98% accuracy
- **Table Tennis**: An Energy Peak Detection based lightweight approach achieved an F-score of 81%
- **Badminton**: An algorithm for badminton serve detection achieved 95% accuracy

On-Court Validation

The gap between machine learning algorithm accuracy (>95%) and real-world on-court performance (85%) highlights the challenges of detecting ball impacts in actual playing conditions with background noise and environmental variations.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:35:02.056Z)

I have enough to evaluate the claim rigorously. Let me assess it against the five checklist points.

Analysis of the claim: "The acoustic ML classifier achieves >95% detection accuracy, but the end-to-end on-court system drops to 85% average accuracy, showing real-world deployment degrades audio-only hit detection substantially."

The numbers (>95% classifier, 85% whole system on-court) are verbatim accurate from the primary source (PMC11843912). However, there are notable framing issues:

1. **Sport mismatch**: This is a **tennis** study, not badminton. The badminton context is the

user's product. Tennis acoustics (ball-on-string, ground bounce) differ materially from badminton (shuttle "thwack").

2. **"Audio-only hit detection" mischaracterizes the 85%: The 85% is the "whole system" which includes peak detection + ML classification + timing calculation**, and the system measures BOTH racket impacts AND **ground rebounds/bounces** (badminton has no ground bounce). Rebound detection is what dragged accuracy down (70.83%-86.25%), while impact (hit) detection was 81.25%-96.25%. So "audio-only hit detection drops to 85%" conflates hit detection with rebound detection and timing.

3. **Tiny sample**: 10 total participants, amateurs/semi-pros, not a robust deployment benchmark.

4. **The directional point is broadly true and supported**: lab?field degradation is real and the magnitude (10pp drop) is reasonable and corroborated by the general literature gap.

The claim's **spirit** (real-world degradation of audio detection) is supported, but the **specifics** are an overreach: wrong sport implied for the context, and "audio-only hit detection drops to 85%" misrepresents what the 85% measures (a multi-stage system averaging hits AND ground rebounds AND timing, on tennis). This is a meaningful misread of a strong-sounding number. Per the rubric, I lean refuted because the claim overstates/misreads what the quote supports.

10. user (2026-06-06T17:35:10.941Z)

Structured output provided successfully

11. assistant (2026-06-06T17:35:17.747Z)

I've completed the verification and returned my assessment via StructuredOutput.

Verdict: refuted = true (high confidence)

The claim's raw numbers (>95% classifier, 85% on-court whole system) are quoted accurately, but the claim is an overreach/misread in ways that matter for our badminton context:

- Wrong sport** ? it's a tennis study, not badminton (no shuttle, has ground bounce).
- "Audio-only hit detection drops to 85%" is a misread** ? the 85% is the whole multi-stage system (peak detection + ML + timing) averaging across both racket impacts AND ground rebounds. Hit detection alone was often >90% (81.25-96.25%); rebound detection (70.83-86.25%) is what dragged the average down ? and rebounds don't exist in badminton.
- Weak basis for a strong generalization** ? n=10, amateurs/semi-pros, single action-cam mic; a pilot, not a deployment benchmark.

The directional truth (lab?field acoustic degradation is real, ~10pp) holds, but the claim attributes the drop specifically to "audio-only hit detection" and implicitly transfers a tennis result onto badminton hit detection, which the source does not support.